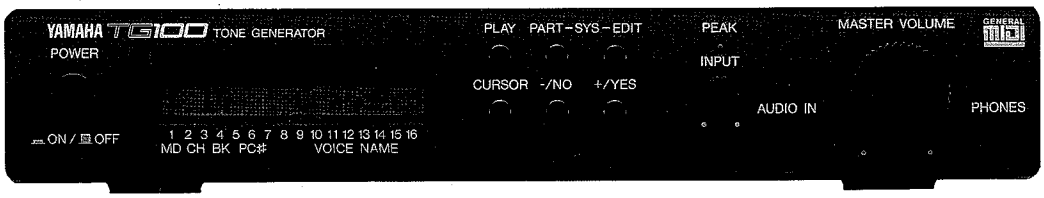
TONE GENERATOR

TG100

SERVICE MANUAL



TG100

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PARTS LIST

LM 011105

YAMAHA CORP.
HAMAMATSU, JAPAN
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IMPORTANT NOTICE

This manual has been provided for the use of authorized Yamaha Retailers and their service personnel. It has been assumed that basic service procedures inherent to the industry, and more specifically Yamaha Products, are already known and understood by the users, and have therefore not been restated.

WARNING: Failure to follow appropriate service and safety procedures when servicing this product may result in personal injury, destruction of expensive components and failure of the product to perform as specified. For these reasons, we advise all Yamaha product owners that all service required should be performed by an authorized Yamaha Retailer or the appointed service representative.

IMPORTANT: The presentation or sale of this manual to any individual or firm does not constitute authorization, certification, recognition of any applicable technical capabilities, or establish a principle-agent relationship of any form.

The data provided is believed to be accurate and applicable to the unit(s) indicated on the cover. The research, engineering, and service departments of Yamaha are continually striving to improve Yamaha products. Modifications are, therefore, inevitable and changes in specification are subject to change without notice or obligation to retrofit. Should any discrepancy appear to exist, please contact the distributor's Service Division.

WARNING: Static discharges can destroy expensive components. Discharge any static electricity your body may have accumulated by grounding yourself to the ground buss in the unit (heavy gauge black wires connect to this buss).

IMPORTANT: Turn the unit OFF during disassembly and parts replacement. Recheck all work before you apply power to the unit.

WARNING: CHEMICAL CONTENT NOTICE!

The solder used in the production of this product contains LEAD. In addition, other electrical/electronic and/or plastic (where applicable) components may also contain traces of chemicals found by the California Health and Welfare Agency (and possibly other entities) to cause cancer and/or birth defects or other reproductive harm.

DO NOT PLACE SOLDER, ELECTRICAL/ELECTRONIC OR PLASTIC COMPONENTS IN YOUR MOUTH FOR ANY REASON WHAT SO EVER!

Avoid prolonged, unprotected contact between solder and your skin! When soldering, do not inhale solder fumes or expose eyes to solder/flux vapor!.

If you come in contact with solder or components located inside the enclosure of this product, wash your hands before handling food.

■ SPECIFICATIONS

Technical specifications

Internal ROM voices	192 instrument voices and 10 drum kits
Internal RAM voices	64 Internal voice
Polyphony	28-note (Dynamically allocated)
Multi-timbral	16 voices simultaneous (voices assigned to 16 Parts)
Sound sampling	AWM (Advanced Wave Memory)
Reverb effect	Yamaha custom DSP (Digital Signal Processor)
Sound module mode	General MIDI LEVEL1
	Disk Orchestra (Yamaha)
	C/M (CM-64 semi-compatible)
Demo song	1 (not editable, stored in ROM)
Controls	MASTER VOLUME, INPUT, CONTRAST
Buttons	PLAY, PART, EDIT, CURSOR, -1/NO, +1/YES
Indicators	PEAK
LCD display	1-line 16-character
Audio connections	
LINE OUT	1/4" (6.35mm) mono jack socket x2
AUDIO IN	3.5mm stereo mini jack x1
PHONES	3.5mm stereo mini jack x1
MIDI connections	IN, OUT THRU (5-PIN DIN socket)
TO HOST	8-PIN mini DIN socket
Host computer selection and data transfer rate	MIDI - 31,250 bps (bits per second)
	Mac - 31,250 bps
	PC-1 - 31,250 bps
	PC-2 - 38,400 bps
Power supply voltage	15V, 500mA
DC IN connection	2.1mm mini power type (for use with PA-1505 adaptor)
Dimensions	220 x 196.5 x 40.6 mm (8.6" x 7.7" x 1.6") W x D x H
Weight	1.0kg
Supplied accessories	PA-1505 power supply adaptor
Optional accessories	RK101 19" rack mounting adaptor

Rack mounting

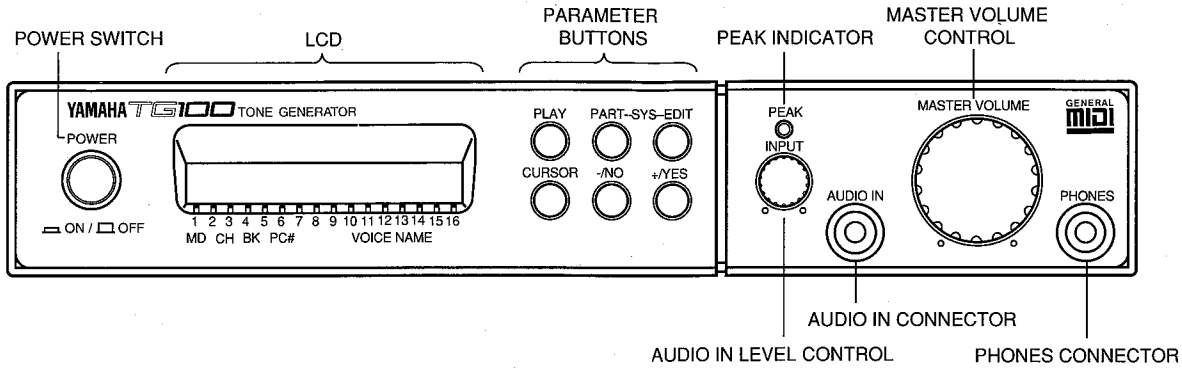
The TG100 can be rack-mounted using one of the "half-rack-size" adaptors that are available (i.e. Yamaha RK101) The TG100 is supplied with two screws for fixing it to an adaptor. Use either these screws, or the screws supplied with the adaptor. Screws must be M3 x 8mm.

■総合仕様

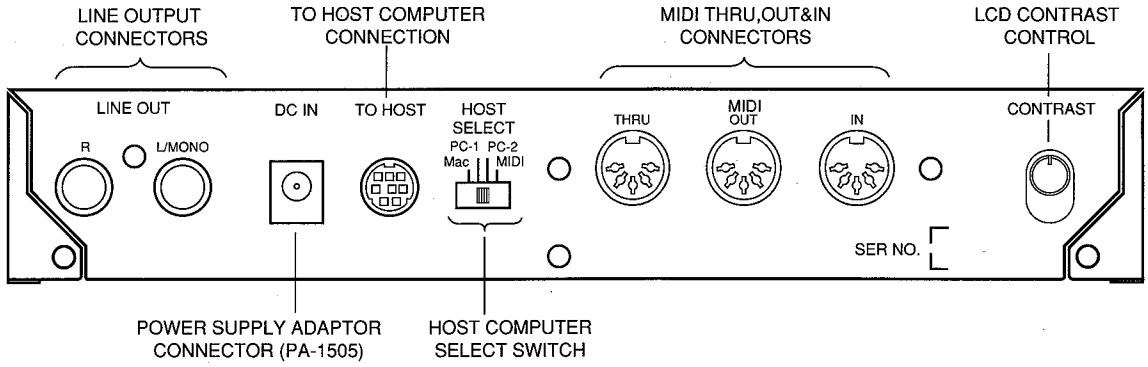
音源方式	AWM音源 リバーブ内蔵 最大同時28音発音、後着優先 最大16音色同時発音
マルチティンバー	16チャンネル DVA 1ボイスAWM×2までのレイヤー可能 ドラムトラックの優先発音
音源機能	GMシステムレベル1規格準拠 MIDI BANK SELECTにより音色バンク変更
インターフェース機能	従来のMIDIシーケンサー、キーボードも接続可能
互換性	GMに対応 ディスクオーケストラに対応 従来のC/Mに対応（一部エクスクルーシブメッセージは除く）
プリセット音色数	202音色（ノーマルボイス192音色、ドラムボイス10音色）
インターナル音色数	64音色
接続端子	
フロント	PHONES×1（ステレオミニジャック） AUDIO IN×1（ステレオミニジャック）
リア	LINE OUT×2（R,L/MONO） MIDI IN MIDI OUT MIDI THRU TO HOST（MINI DIN 8P）
電源電圧	15V 500mA
外形寸法	220(W)×196.5(D)×40.6(H)
重量	1.0kg
付属品	電源アダプター PA-1505　×1 取扱説明書　×1
オプション（別売品）	ラックマウントキット RK101

■ PANEL LAYOUT（パネルレイアウト）

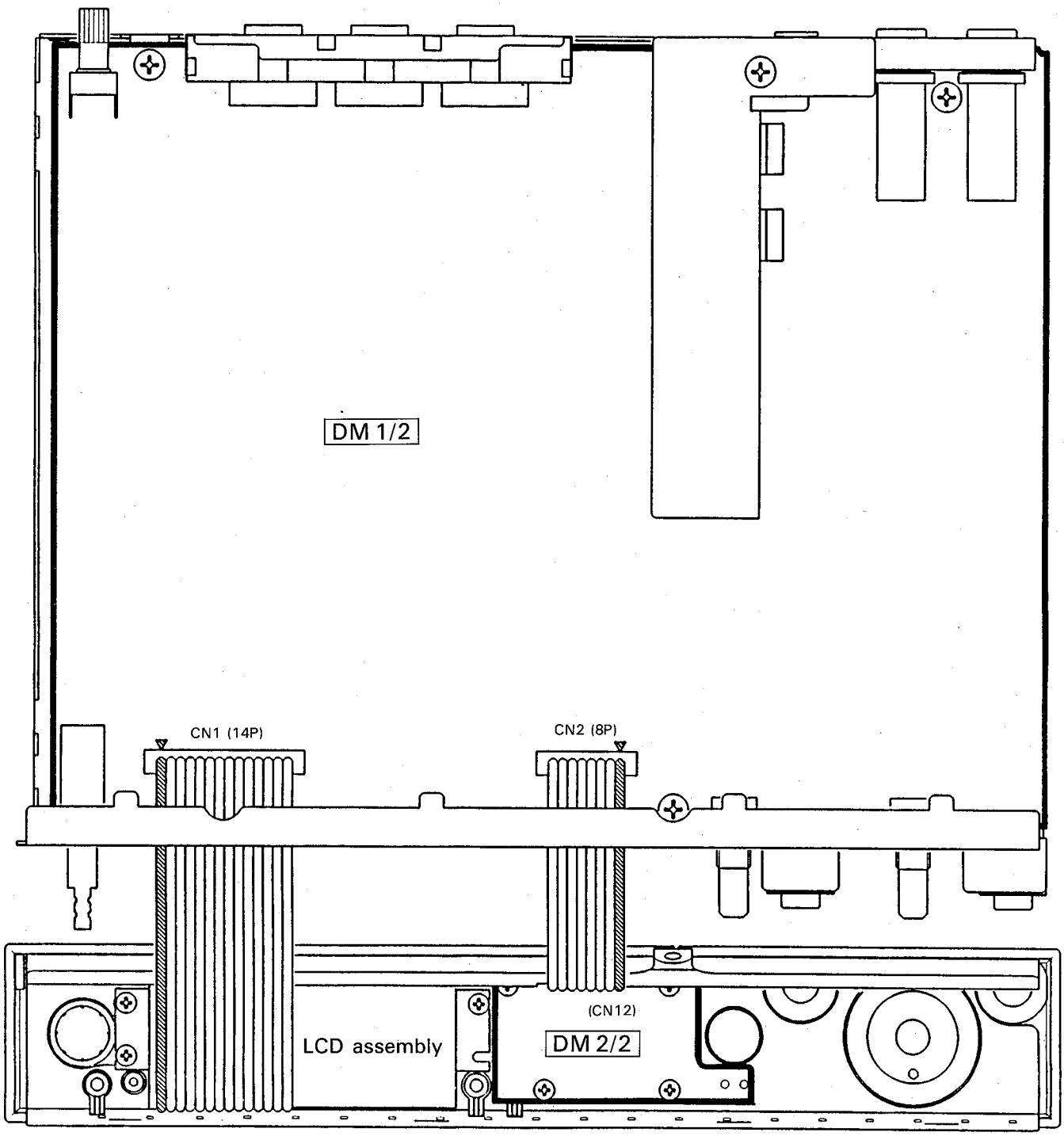
● Front panel



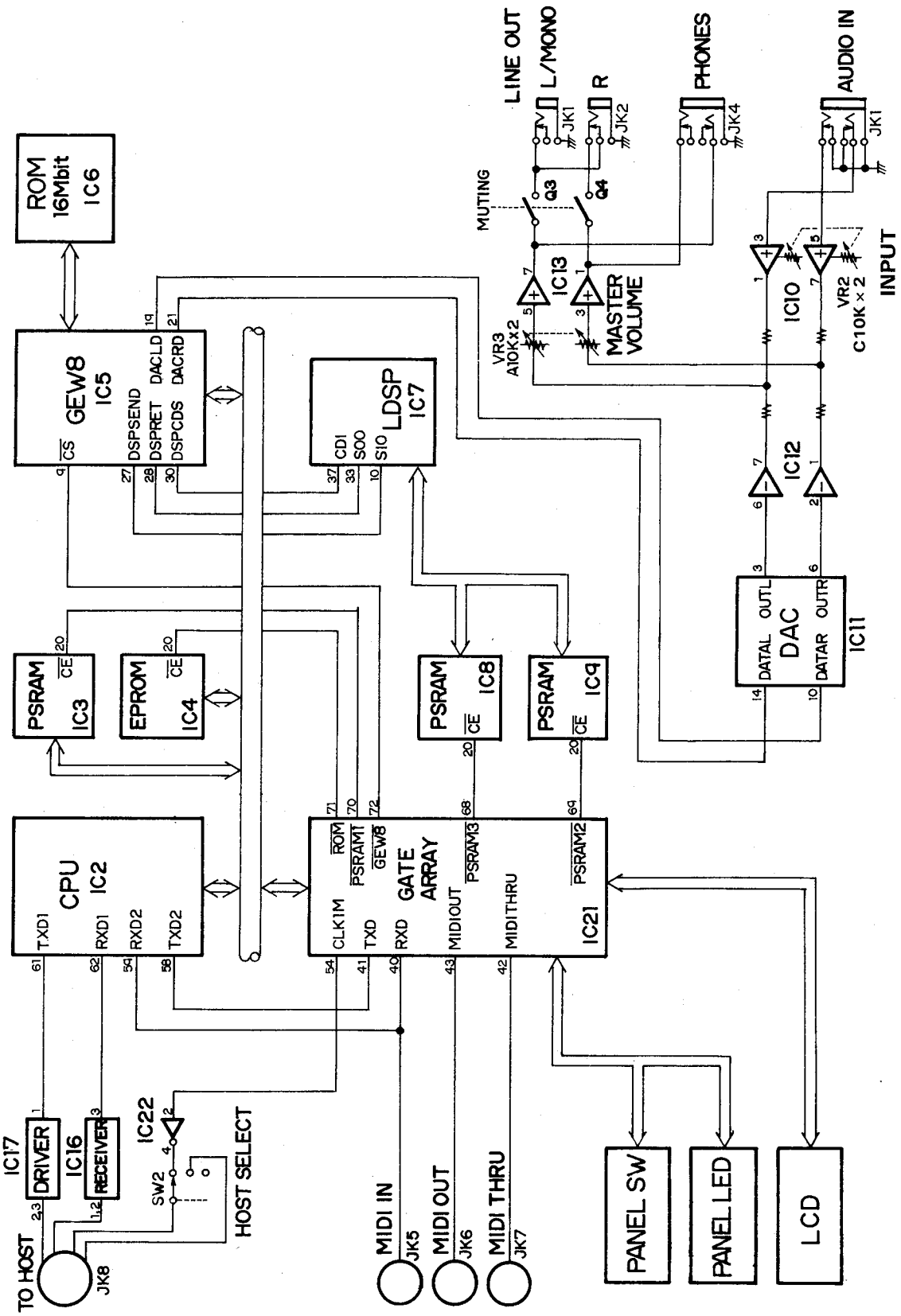
● Rear panel



■ CIRCUIT BOARD LAYOUT（ユニットレイアウト）



■ BLOCK DIAGRAM（ブロックダイアグラム）



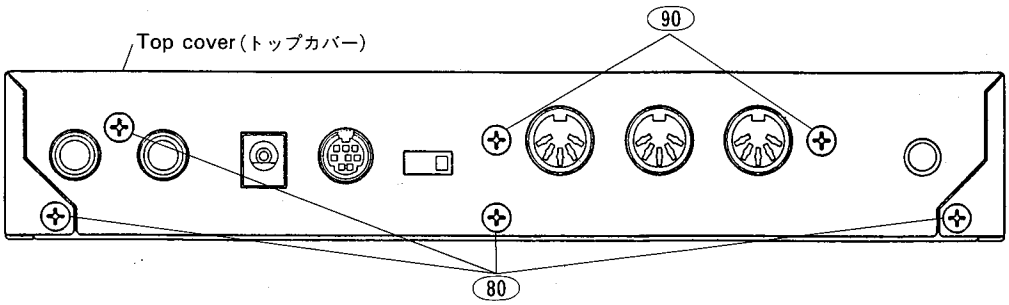
■ DISASSEMBLY PROCEDURE（分解手順）

1. Top Cover Removal

1-1. Remove the four (4) screws marked (80) and two (2) screws marked (90). (Fig. 1)
1-2. Slide the top cover backward, and lift it from the left, then the top cover can be removed.

1．トップカバーの外し方

1-1 (80) のネジ4本と (90) のネジ2本を外し、トップカバーを後側にずらして、左側を持ち上げながら外します。(図1)



(Fig. 1)

(80): Bonding tapping screw-C(ボンディングCタイト) 3.0×6 FCM3BL
(90): Bind head tapping screw-P(＋バインドPタイト) 3.0×8 FCM3BL

2. Front Cover Assembly Removal

2-1. Remove the top cover. (see procedure 1)
2-2. Pull off the three (3) knobs marked (100), (110) and (120). (Fig. 2)
2-3. Pull off the ribbon cables, then remove the two (2) screws marked (30) and two (2) screws marked (50a). (Fig. 2)
2-4. Remove the hexagonal nut marked (60), then the front cover assembly can be removed. (Fig. 2 and Fig. 3)
2-5. Remove the two (2) screws marked (20a), then the DM2/2 circuit board can be removed. (Fig. 3 and Fig. 4)
* The DM2/2 circuit board is not a part of the front panel assembly.

2．フロントカバーAss'yの外し方

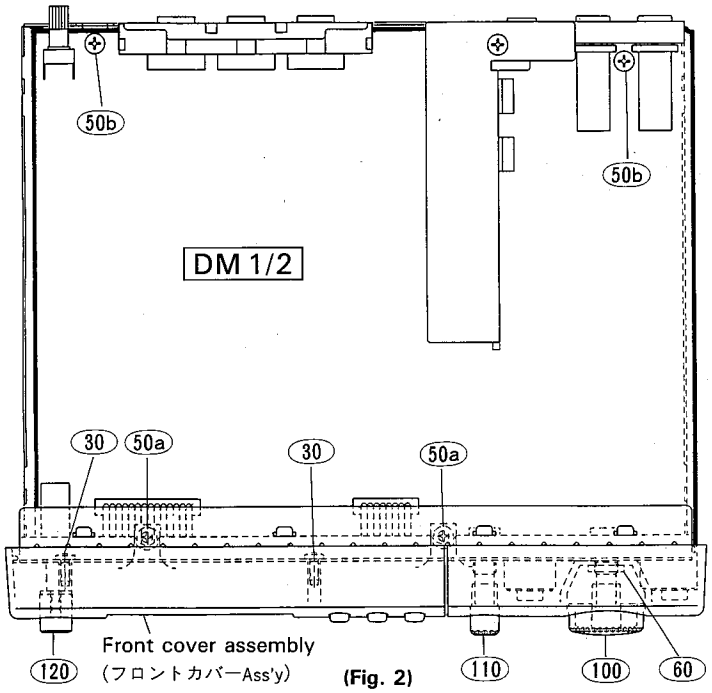
2-1 トップカバーを外します。(1項参照)
2-2 (100) のボリュームツマミと (110) のインプットツマミと (120) のプッシュボタンを引き抜きます。(図2)
2-3 最初に束線を外します。次に (30) のネジ2本と (50a) のネジ2本を外します。(図2)
2-4 (60) の特殊六角ナットを外し、フロントカバーAss'yの右側を手前に引きながら外します。(図2、図3)
2-5 (20a) のネジ4本を外し、DM2/2シートを外します。(図3、図4)
＊DM2/2シートは、フロントカバーAss'yの構成部品ではありません。

3. DM1/2 Circuit Board Removal

3-1. Remove the top cover. (see procedure 1)
3-2. Remove the front cover assembly. (see procedure 2)
3-3. After the two (2) screws marked (50b) have been removed, then the DM1/2 circuit board can be removed. (Fig. 2)

3．DM1/2シートの外し方

3-1 トップカバーを外します。(1項参照)
3-2 フロントカバーAss'yを外します。(2項参照)
3-3 (50b) のネジ2本を外し、ボトムカバーよりDM1/2シートを外します。(図2)



(Fig. 2)

(30) : Bind head tapping screw-P(+バインドPタイト) 3.0×8 ZMC2BL
(50) : Bind head tapping screw-C(+バインドCタイト) 3.0×6 ZMC2BL
(60) : Hexagonal Nut(特殊六角ナット) 7.0 ZMC2BL

4. DM2/2 Circuit Board Removal

4-1. Remove the top cover. (see procedure 1)
4-2. Remove the front cover assembly. (see procedure 2)
4-3. Remove the two (2) screws marked (20a), then the DM2/2 circuit board can be removed. (Fig. 3 and Fig. 4)
4-4. Pull out the six (6) OP. buttons from the DM2/2 circuit board. (Fig. 4)

5. LCD Assembly Removal

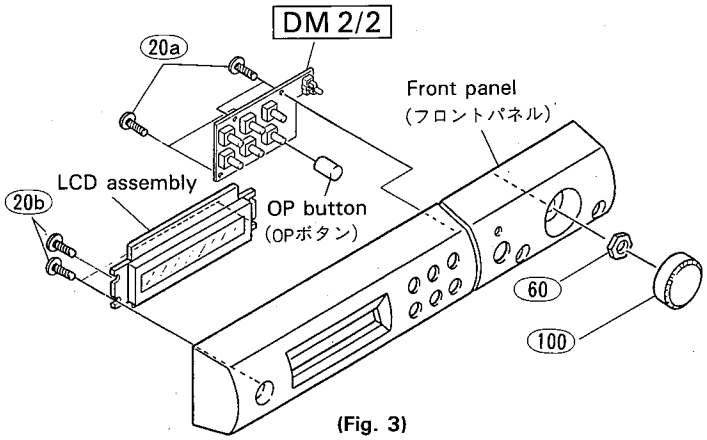
5-1. Remove the top cover. (see procedure 1)
5-2. Remove the front cover assembly. (see procedure 2)
5-3. Remove the three (3) screws marked (20b), then the LCD assembly can be removed. (Fig. 3 and Fig. 4)

4．DM2/2シートの外し方

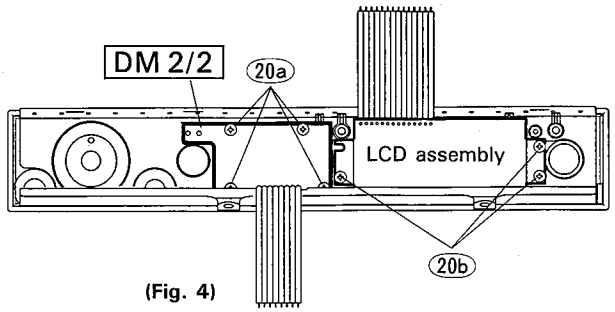
4-1 トップカバーを外します。(1項参照)
4-2 フロントカバーAss'yを外します。(2項参照)
4-3 (20a) のネジ4本を外し、DM2/2シートを外します。(図3、図4)
4-4 DM2/2シートより、OPボタン6個を外します。(図4)

5．LCD Ass'yの外し方

5-1 トップカバーを外します。(1項参照)
5-2 フロントカバーAss'yを外します。(2項参照)
5-3 (20b) のネジ3本を外し、LCD Ass'yを外します。(図3、図4)



(Fig. 3)



(Fig. 4)

(20) : Bind head tapping screw-P (+バインドPタイト) 2.0×4 ZMC2BL

(20) : Bind head tapping screw-P (+バインドPタイト) 2.0×4 ZMC2BL
(60) : Hexagonal Nut(特殊六角ナット) 7.0 ZMC2BL

■LSI PIN DESCRIPTION (LSI端子機能表)

• HD6435208A00P (XK278A00) CPU <H8/520>

Pin No.	Name	I/O	Function	Pin No.	Name	I/O	Function
1	EXT	I	Clock	33	A7	O	Address bus
2	XTAL	I	Clock	34	A8	O	Address bus
3	$\overline{WAIT}$	I	Bus cycle wait	35	A9	O	Address bus
4	$\overline{IRQ0}$	O	Interrupt request	36	A10	O	Address bus
5	A18	O	Address bus	37	A11	O	Address bus
6	A17	O	Address bus	38	A12	O	Address bus
7	A16	O	Address bus	39	A13	O	Address bus
8	$\overline{AS}$	O	Address strobe	40	A14	O	Address bus
9	$\overline{RD}$	O	Read control	41	A15	O	Address bus
10	$\overline{WR}$	O	Write control	42	VCC		Power supply
11	VCC		Power supply	43	P50	O	Port 5
12	MD0	I	Mode select	44	P51	O	Port 5
13	MD1	I	Mode select	45	P52	O	Port 5
14	MD2	I	Mode select	46	P53	O	Port 5
15	$\overline{RES}$	I	Reset	47	P54	O	Port 5
16	NM1	I	Non-maskable interrupt	48	P55	O	Port 5
17	VSS		Ground	49	P56	O	Port 5
18	D0	I/O	Data bus	50	P57	O	Port 5
19	D1	I/O	Data bus	51	VSS		Ground
20	D2	I/O	Data bus	52	AVSS		Analog ground
21	D3	I/O	Data bus	53	AN0	I	Analog data input
22	D4	I/O	Data bus	54	AN1	I	Analog data input
23	D5	I/O	Data bus	55	AN2	I	Analog data input
24	D6	I/O	Data bus	56	AN3	I	Analog data input
25	D7	I/O	Data bus	57	AVCC		Analog power supply
26	A0	O	Address bus	58	TXD2	O	Transmit data
27	A1	O	Address bus	59	RXD2	I	Receive data
28	A2	O	Address bus	60	A19	O	Address bus
29	A3	O	Address bus	61	TXD1		Transmit data
30	A4	O	Address bus	62	RXD1	I	Receive data
31	A5	O	Address bus	63	SCLK	I	Clock for serial operation
32	A6	O	Address bus	64	Vss		Ground

• YM3413 (XE449A00) LDSP (Digital Signal Proccesor)

PIN NO.	NAME	I/O	FUNCTION	PIN NO.	NAME	I/O	FUNCTION
1	VDD		DC supply (+5V)	21	A5	O	Address bus
2	D7	I/O	Data bus	22	A6	O	Address bus
3	D6	I/O	Data bus	23	A7	O	Address bus
4	D5	I/O	Data bus	24	A8	O	Address bus
5	D4	I/O	Data bus	25	A9	O	Address bus
6	D3	I/O	Data bus	26	A10	O	Address bus
7	D2	I/O	Data bus	27	A11	O	Address bus
8	D1	I/O	Data bus	28	A12	O	Address bus
9	D0	I/O	Data bus	29	A13	O	Address bus
10	SI0	I	Serial data input	30	A14	O	Address bus
11	SI1	I	Serial data input	31	A15	O	Address bus
12	SYW	I	Sync pulse	32	A16	O	Address bus
13	$\overline{WE}$	O	Write enable	33	SO0	O	Serial data output
14	$\overline{OE}$	O	Output enable	34	XCLK	I	Clock
15	A0	O	Address bus	35	$\overline{IC}$	I	Initial Clear
16	A1	O	Address bus	36	$\overline{CRS}$	I	CD counter reset
17	A2	O	Address bus	37	CDI	I	CD input
18	A3	O	Address bus	38	CDo	O	CD output
19	A4	O	Address bus	39	SO1	O	Serial data output
20	Vss		Ground	40	CLK	I	Clock

• YMW-258-F (XJ427A00) GEW8 (PCM Tone Generator)

PIN NO.	NAME	I/O	FUNCTION	PIN NO.	NAME	I/O	FUNCTION
1	NC			41	Vss		Ground
2	Vss		Ground	42	NC		
3	D0	I/O	CPU data bus	43	Vss		Ground
4	D7	I/O		44	AB0	O	Voice memory address bus
5	A0	I	CPU address bus	45	DB7	I/O	Voice memory data bus
6	A1	I		46	AB1	O	
7	A2	I		47	AB2	O	
8	A3	I		48	AB10	O	
9	$\overline{CS}$	I	Chip select	49	AB3	O	
10	$\overline{RD}$	I	Read control	50	AB4	O	
11	$\overline{WR}$	I	Write control	51	AB11	O	
12	XIN		Clock	52	AB5	O	
13	XOUT			53	AB9	O	
14	$\overline{IC}$	I	Initial clear	54	AB6	O	Voice memory address bus
15	TST0		Test pin	55	AB8	O	
16	TST1			56	AB7	O	
17	$\overline{Vss}$		Ground	57	AB13	O	
18	$\overline{DITHER}$		Not used	58	AB12	O	
19	DACLD	O	Data output, L channel	59	AB14	O	
20	DACDCLK	O	Bit clock output to DAC	60	AB15	O	
21	DACRD	O	Data output, R channel	61	AB17	O	
22	DACLE	O	Word clock output to DAC	62	Vss		Ground
23	NC			63	Vss		
24	DACMC	O	System clock output to DAC	64	Vss		
25	CH27		Not used	65	Vss		
26	DSPSYW			66	AB16	O	Voice memory address bus
27	DSPSEND			67	AB18	O	
28	$\overline{DSPRET}$			68	AB19	O	
29	$\overline{DSPIC}$			69	AB20	O	
30	DSPCDS			70	AB21	O	
31	DSPCLK			71	MRD(MWR)	O	Memory read control
32	VDD		Power supply	72	VDD		Power supply
33	DB3	I/O	Voice memory data bus	73	MWR(MRD)	O	Memory write control
34	DB2	I/O		74	D3	I/O	CPU data bus
35	DB4	I/O		75	D4	I/O	
36	DB1	I/O		76	D2	I/O	
37	DB5	I/O		77	D5	I/O	
38	DB0	I/O		78	D1	I/O	
39	DB6	I/O		79	D6	I/O	
40	NC			80	Vss		Ground

• PCM69P-C (XJ429A00) DAC (Digital to Analog Converter)

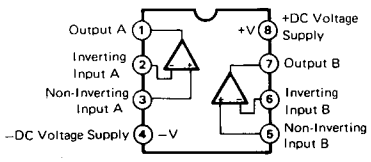
PIN NO.	NAME	I/O	FUNCTION	PIN NO.	NAME	I/O	FUNCTION
1	+Vcc		Analog power supply	9	OGND		Digital ground
2	VC, L		V-common, L channel	10	DA, R		Data input, R channel
3	IO, L		Current output, L channel	11	BCK		Bit clock
4	SER		Servo filter	12	CLK		System clock
5	REF		Reference filter	13	WDCK		Word clock
6	IO, R		Current output, R channel	14	DA, L		Data input, L channel
7	VC, R		V-common, R channel	15	TP1		Test pin
8	AGND		Analog ground	16	+VDD		Digital power supply

• **HG62E11R54FS** (XK462A00) Gate Array

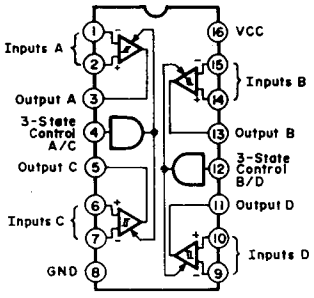
PIN NO.	NAME	I/O	FUNCTION	PIN NO.	NAME	I/O	FUNCTION
1	DLCD0	I/O	LCD data bus	41	TXD	I	
2	DLCD1	I/O		42	MIDITHRU	O	MIDI data thru.
3	DLCD2	I/O		43	MIDIOUT	O	MIDI data output
4	DLCD3	I/O		44	D0	I/O	CUP data bus
5	DLCD4	I/O		45	D1	I/O	
6	DLCD5	I/O		46	D2	I/O	
7	DLCD6	I/O		47	D3	I/O	
8	DLCD7	I/O		48	D4	I/O	
9	LED0	O	LED drive	49	D5	I/O	
10	LED1	O		50	D6	I/O	
11	LED2	O		51	D7	I/O	
12	(GND)			52	(GND)		
13	LED3	O		53	MCLK	I	Master clock
14	LED4	O		54	CLK1M	O	Clock 1MHz
15	LED5	O		55	$\overline{WAIT}$	O	Wait control
16	LED6	O		56	$\overline{AS}$	I	Address strobe
17	LED7	O		57	$\overline{RD}$	I	Read control
18	LED8	O		58	$\overline{WR}$	I	Write control
19	(GND)			59	A19	I	CPU address bus
20	LED9	O		60	A18	I	
21	LED10	O		61	A14	I	
22	LED11	O		62	A13	I	
23	LED12	O		63	A12	I	
24	LED13	O		64	A0	I	
25	LED14	O		65	DSPA15	I	DSP address
26	(GND)			66	DSPCLK	I	DSP clock
27	LED15	O		67	(GND)		
28	LED16	O		68	$\overline{PSRAM3}$	O	PSRAM chip enable
29	LED17	O		69	$\overline{PSRAM2}$	O	
30	LED18	O		70	$\overline{PSRAM1}$	O	
31	SW0	I	Switch scan	71	$\overline{ROM}$	O	ROM chip enable
32	SW1	I		72	$\overline{GEW8}$	O	GEW8 chip enable
33	(VCC)	I		73	(VCC)		
34	SW2	I		74	$\overline{OE/RFSH}$	O	Output enable/Refresh
35	SW3	I		75	$\overline{RESOUT}$	O	Reset output
36	SW4	I		76	RESIN	I	Reset input
37	SW5	I		77	(GND)		
38	SW6	I		78	RS	O	LCD register select
39	(GND)			79	LCDR/$\overline{W}$	O	LCD read/write
40	RXD	I	MIDI data input	80	LCDE	O	LCD enable

■ IC BLOCK DIAGRAM (ICブロック図)

• **RC4558DV** (IG001390)
• **NJM4556** (IG042500)
Dual Operational Amplifier



• **MC3486** (XB016A00)
Quad Line Receiver with 3-State Outputs



• **MC3487** (XB015A00)
Quad Line Driver with 3-State Outputs

